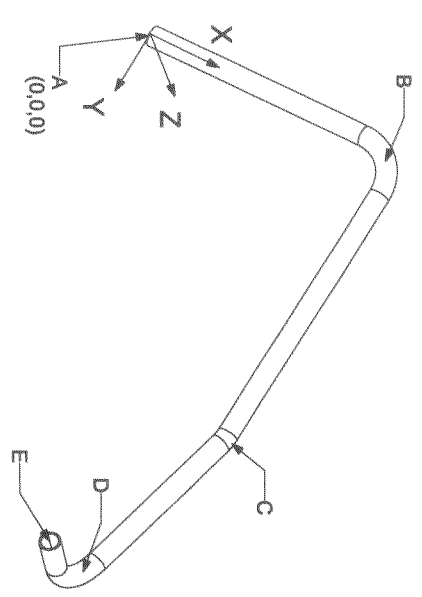
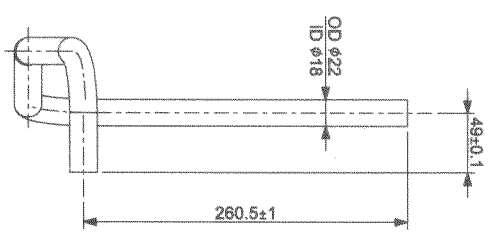
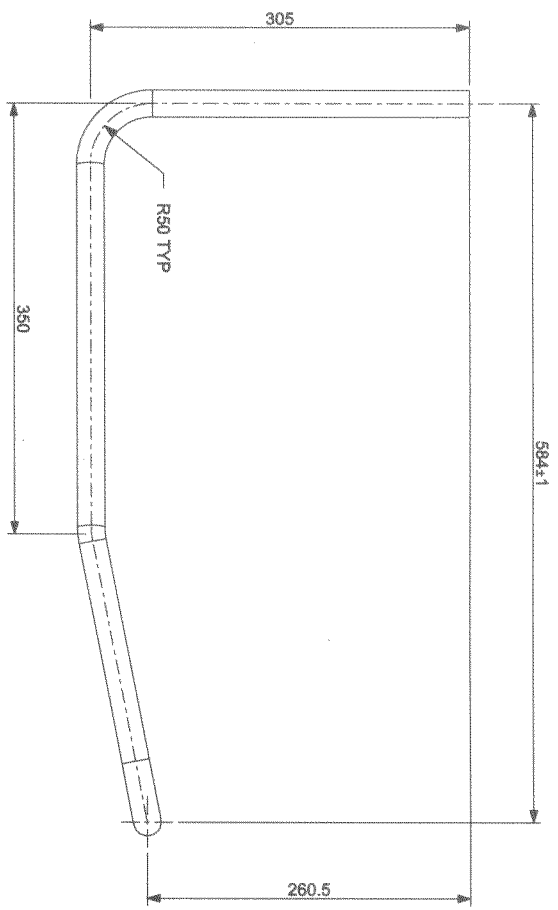


| POINT | COORDINATE X' | COORDINATE Y' | COORDINATE Z' | BEND RADIUS |
|-------|---------------|---------------|---------------|-------------|
| A     | 0             | 0             | 0             | 50          |
| B     | 305           | 0             | 0             | 50          |
| C     | 305           | 350           | 50            | 50          |
| D     | 260.5         | 584           | 50            | 50          |
| E     | 260.5         | 584           | -49           |             |

| 01 PIPE DESIGN CHANGED |                   |       |      |      |      |      |      |      |      |            |      |      |        |               |
|------------------------|-------------------|-------|------|------|------|------|------|------|------|------------|------|------|--------|---------------|
| Alt.                   |                   |       |      |      |      |      |      |      |      | Amendments |      |      |        |               |
| Sur. Sym.              | N1                | N2    | N3   | N4   | N5   | N6   | N7   | N8   | N9   | N10        | N11  | N12  | Mat.:- | Sur. Treat.:- |
| Re Value               | µm                | 0.025 | 0.05 | 0.1  | 0.2  | 0.4  | 0.8  | 1.6  | 3.2  | 6.3        | 12.5 | 25   | 50     |               |
| IS:3073                | REF. NO.:-        |       |      |      |      |      |      |      |      |            |      |      |        |               |
| Gen. Tol. IS:2102      | Fin. Tol. IS:2102 | 2016  | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016       | 2016 | 2016 | 2016   | 2016          |
| 0.5-3                  | 0.05-0.1          | 0.05  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1        | 0.1  | 0.1  | 0.1    | 0.1           |
| >3-6                   | 0.05-0.1          | 0.05  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1        | 0.1  | 0.1  | 0.1    | 0.1           |
| >6-30                  | 0.1-0.2           | 0.1   | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2        | 0.2  | 0.2  | 0.2    | 0.2           |
| >30-120                | 0.15-0.3          | 0.15  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3        | 0.3  | 0.3  | 0.3    | 0.3           |
| >120-400               | 0.2-0.5           | 0.2   | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5        | 0.5  | 0.5  | 0.5    | 0.5           |
| >400-1000              | 0.3-0.8           | 0.3   | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8        | 0.8  | 0.8  | 0.8    | 0.8           |
| >1000-2500             | 0.5-1.2           | 0.5   | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2        | 1.2  | 1.2  | 1.2    | 1.2           |
| >2500-4000             | 0.8-1.6           | 0.8   | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6        | 1.6  | 1.6  | 1.6    | 1.6           |
| 0.5-3                  | 0.05-0.1          | 0.05  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1        | 0.1  | 0.1  | 0.1    | 0.1           |
| >3-6                   | 0.05-0.1          | 0.05  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1        | 0.1  | 0.1  | 0.1    | 0.1           |
| >6-30                  | 0.1-0.2           | 0.1   | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2        | 0.2  | 0.2  | 0.2    | 0.2           |
| >30-120                | 0.15-0.3          | 0.15  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3        | 0.3  | 0.3  | 0.3    | 0.3           |
| >120-400               | 0.2-0.5           | 0.2   | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5        | 0.5  | 0.5  | 0.5    | 0.5           |
| >400-1000              | 0.3-0.8           | 0.3   | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8        | 0.8  | 0.8  | 0.8    | 0.8           |
| >1000-2500             | 0.5-1.2           | 0.5   | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2        | 1.2  | 1.2  | 1.2    | 1.2           |
| >2500-4000             | 0.8-1.6           | 0.8   | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6        | 1.6  | 1.6  | 1.6    | 1.6           |

| Gen. Tol. IS:2102 | Fin. Tol. IS:2102 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 |
|-------------------|-------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 0.5-3             | 0.05-0.1          | 0.05 | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  |
| >3-6              | 0.05-0.1          | 0.05 | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  |
| >6-30             | 0.1-0.2           | 0.1  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  |
| >30-120           | 0.15-0.3          | 0.15 | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  |
| >120-400          | 0.2-0.5           | 0.2  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  |
| >400-1000         | 0.3-0.8           | 0.3  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  |
| >1000-2500        | 0.5-1.2           | 0.5  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  |
| >2500-4000        | 0.8-1.6           | 0.8  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  |

| Gen. Tol. IS:2102 | Fin. Tol. IS:2102 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 |
|-------------------|-------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 0.5-3             | 0.05-0.1          | 0.05 | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  |
| >3-6              | 0.05-0.1          | 0.05 | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  |
| >6-30             | 0.1-0.2           | 0.1  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  |
| >30-120           | 0.15-0.3          | 0.15 | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  |
| >120-400          | 0.2-0.5           | 0.2  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  |
| >400-1000         | 0.3-0.8           | 0.3  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  |
| >1000-2500        | 0.5-1.2           | 0.5  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  |
| >2500-4000        | 0.8-1.6           | 0.8  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  |

NOTE:-  
1. UNSPECIFIED TOLERANCE AS PER IS:2102-COARSE  
2. REMOVE SHARP EDGES AND BURRS.

| Gen. Tol. IS:2102 | Fin. Tol. IS:2102 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 |
|-------------------|-------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 0.5-3             | 0.05-0.1          | 0.05 | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  |
| >3-6              | 0.05-0.1          | 0.05 | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  |
| >6-30             | 0.1-0.2           | 0.1  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  |
| >30-120           | 0.15-0.3          | 0.15 | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  |
| >120-400          | 0.2-0.5           | 0.2  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  |
| >400-1000         | 0.3-0.8           | 0.3  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  |
| >1000-2500        | 0.5-1.2           | 0.5  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  |
| >2500-4000        | 0.8-1.6           | 0.8  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  |

| Gen. Tol. IS:2102 | Fin. Tol. IS:2102 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 |
|-------------------|-------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 0.5-3             | 0.05-0.1          | 0.05 | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  |
| >3-6              | 0.05-0.1          | 0.05 | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  |
| >6-30             | 0.1-0.2           | 0.1  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  |
| >30-120           | 0.15-0.3          | 0.15 | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  |
| >120-400          | 0.2-0.5           | 0.2  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  |
| >400-1000         | 0.3-0.8           | 0.3  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  |
| >1000-2500        | 0.5-1.2           | 0.5  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  |
| >2500-4000        | 0.8-1.6           | 0.8  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  |

| Gen. Tol. IS:2102 | Fin. Tol. IS:2102 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 |
|-------------------|-------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 0.5-3             | 0.05-0.1          | 0.05 | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  |
| >3-6              | 0.05-0.1          | 0.05 | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  |
| >6-30             | 0.1-0.2           | 0.1  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  |
| >30-120           | 0.15-0.3          | 0.15 | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  |
| >120-400          | 0.2-0.5           | 0.2  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  |
| >400-1000         | 0.3-0.8           | 0.3  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  |
| >1000-2500        | 0.5-1.2           | 0.5  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  |
| >2500-4000        | 0.8-1.6           | 0.8  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  |

| Gen. Tol. IS:2102 | Fin. Tol. IS:2102 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 | 2016 |
|-------------------|-------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 0.5-3             | 0.05-0.1          | 0.05 | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  |
| >3-6              | 0.05-0.1          | 0.05 | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  | 0.1  |
| >6-30             | 0.1-0.2           | 0.1  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  | 0.2  |
| >30-120           | 0.15-0.3          | 0.15 | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  |
| >120-400          | 0.2-0.5           | 0.2  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  | 0.5  |
| >400-1000         | 0.3-0.8           | 0.3  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  | 0.8  |
| >1000-2500        | 0.5-1.2           | 0.5  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  | 1.2  |
| >2500-4000        | 0.8-1.6           | 0.8  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  | 1.6  |

|                     |                      |           |      |    |    |
|---------------------|----------------------|-----------|------|----|----|
| Project :-<br>EMCBS | >1000-2000           | 10.5      | 11.2 | 13 | 16 |
|                     | >2000-4001           | -         | 12   | 14 | 18 |
|                     | AR CONNECTION PIPE 2 |           |      |    |    |
|                     | 1J106010067          |           |      |    |    |
|                     | Sheet 1 of 1         | Alt no 01 |      |    |    |